

Electrochemical equilibrium

After midterm · Session 6 of 6 · 180 minutes

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The question for today

What determines the sign of a concentration-cell potential?

Before class: 3-page reading · 15–20 minutes

Bring the three preparation answers. Revisit your prediction after one controlled comparison and an independent check.

The 180-minute class

ACTIVITY	MINUTES
Opening question and essential concepts	30
One hand calculation	20
One lab demonstration	20
Break	10
Student pair investigation	70
Discussion and exit question	30

The 70-minute investigation is student work with instructor support.

Two reduction half-reactions

Enter both half-reactions as reductions. Reverse the left reaction and add the right after matching electron counts.

$$E_{cell}^{\circ} = E_{right}^{\circ} - E_{left}^{\circ}$$

Both potentials need the same reference electrode and temperature. Multiplying a half-reaction does not multiply its potential.

Nernst equation from reaction thermodynamics

Combine $\Delta_r G = \Delta_r G^\circ + RT \ln Q$ with $\Delta_r G = -nFE$:

$$E = E^\circ - \frac{RT}{nF} \ln Q$$

At 298.15 K,

$$E = E^\circ - \frac{0.05915935 \text{ V}}{n} \log_{10} Q$$

Changing T requires valid E° data at the new T.

A concentration-cell calculation

At 298.15 K, $n=2$, $a_L=0.01$ and $a_R=1$:

$$Q = 0.01, \quad E = +0.05915935 \text{ V}$$

$E^\circ=0$, $K=1$, but E is nonzero. Reversing the concentrations changes the sign of E . Equal activities give $E=0$.

$\Delta_r G \approx -11.416 \text{ kJ}$ per mol of the written reaction.

An independent check

Use dimensionless activities. Concentration is an approximation with a stated standard concentration and activity coefficient.

Use $E_{\text{right}} - E_{\text{left}}$ and $\Delta G = -nFE$ with the balanced electron count. Equal activities must give zero potential for the same reduction couple.

One demonstration • 20 minutes

Open the core lab

Load the Lab 13 concentration-cell preset. Predict the sign, exchange left and right activities, then set them equal. Keep T and reference conventions fixed.

Keep one saved baseline for the pair investigation.

Pair investigation • 70 minutes

- 10 min: predict and record the baseline.
- 25 min: change one input and calculate.
- 20 min: exchange roles and check the result by hand.
- 15 min: prepare one plot or table and a short explanation.

In Lab 13, compare a concentration cell with the activities exchanged. Predict and explain the sign of E , then independently check $\Delta G = -nFE$.

Discussion and exit question • 30 minutes

Use 20 minutes to compare results and discuss unexpected behavior.

What determines the sign of a concentration-cell potential?

Use the final 10 minutes to explain your answer individually and name one assumption that limits it.

Follow-up practice • 30–40 minutes

In Lab 13, compare a concentration cell with the activities exchanged. Predict and explain the sign of E , then independently check $\Delta G = -nFE$.

Assessment 6 and independent checkpoint

Retain one study JSON, one comparison and a short explanation. Continue from your classroom case. The hand checkpoint is included. With pre-class reading, independent work totals 45–60 minutes.

Optional study

Choose one extension: scale the balanced reaction and electron count, examine activity corrections within the supported model, or explain why equilibrium voltage differs from voltage under load.

[Session appendix](#) · [Module reference](#)

[Learning path](#) · [Offline package](#)

Optional work is additional. Complete the required investigation first.